

Varun Verma

Principal Software Engineer | Backend & Distributed Systems | Cloud Architecture
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EXECUTIVE SUMMARY

Principal-level software engineer and technical lead with 14+ years of experience architecting and delivering enterprise backend platforms, distributed systems, event-driven services, cloud-native infrastructure, and developer platforms across Azure, AWS, and GCP. Deep hands-on expertise in .NET/C#, Python, Go, APIs, microservices, Kafka, data platforms, Kubernetes, observability, identity, and Infrastructure as Code. Experienced in owning architecture from system design and API contracts through reliability, security, deployment, disaster recovery, and operational excellence. Increasingly applying AI/ML and LLM capabilities to backend platforms and engineering automation, while keeping distributed backend architecture, scalability, and platform engineering as the primary focus.

CORE ARCHITECTURE EXPERTISE

Backend Architecture: Distributed systems, microservices, modular monoliths, REST APIs, event-driven architecture, asynchronous processing, integration platforms, API gateways, SDK design, service-to-service communication

Distributed Systems: Apache Kafka, Confluent Cloud, consumer groups, partitioning, retries, idempotency, fault tolerance, disaster recovery, active/active and failover patterns, high availability

Cloud & Platform Engineering: Azure, AWS, GCP, Kubernetes, Docker, Terraform, Bicep, ARM, CloudFormation, GitOps, platform-as-a-product, reusable infrastructure patterns

Reliability & Operations: 99.9%+ availability, observability, OpenTelemetry, Prometheus, Grafana, distributed tracing, structured logging, performance engineering, incident response, SRE practices

Security & Identity: OAuth2, JWT, SAML 2.0, SSO, Azure AD, MFA, TLS, RBAC, enterprise identity and access-control architecture

Data & Integration: SQL Server, PostgreSQL, Oracle, MongoDB, Cosmos DB, DynamoDB, Redis, Elasticsearch, Spark, Databricks, Airflow, Azure Data Factory

AI/ML & LLM Engineering: SageMaker, Azure ML, ML pipelines, RAG, embeddings, LangChain, LangGraph, AI agents, MCP, model observability and AI-assisted automation

PROFESSIONAL EXPERIENCE

Shell, Bangalore

April 2022 – Present

Senior Technical Lead

- Provide technical leadership for enterprise backend, event-streaming, cloud platform, and AI/ML engineering initiatives, driving architecture, design reviews, implementation strategy, reliability, security, and operational readiness.
- Architected and developed .NET and Python SDKs for Confluent Kafka APIs with OAuth2 authentication, simplifying enterprise integration and supporting 10M+ real-time events daily with disaster-recovery capabilities.
- Designed backend microservices secured with enterprise SSO using SAML 2.0 and OAuth2, supporting 1,000+ users with 99.99% availability.
- Designed event-driven integration and data pipelines using Apache Kafka and Apache Airflow, including edge-to-cloud ingestion patterns and resilient processing workflows.
- Engineered automated multi-account AWS provisioning using Infrastructure as Code, reducing environment setup from days to minutes across 50+ accounts.
- Reduced operational toil by 97%, from approximately 9 hours to 5 minutes, through automated infrastructure provisioning and delivery using Azure DevOps and GitHub Actions.
- Architected production MLOps and LLMops pipelines using AWS SageMaker and Apache Airflow, integrating model training, validation, deployment, and operational controls with 99.9% uptime SLA.
- Built ML-powered document-processing services using AWS Textract and custom NLP models, processing 100K+ documents monthly with 94% extraction accuracy.
- Designed agentic AI workflows using LangGraph and LangChain for incident response, root-cause analysis, and infrastructure remediation, reducing mean time to resolution by 60%.
- Implemented production observability using OpenTelemetry and Prometheus across backend and AI workloads, covering latency, resource utilization, traces, and model-specific operational signals.

Deloitte, Gurgaon

February 2021 – April 2022

Manager (Architect)

- Led solution architecture for enterprise applications, with emphasis on backend service decomposition, integration, security, cloud architecture, and automation.
- Designed AI-enhanced microservices architecture for a fintech platform, integrating ML-based fraud detection with real-time payment-processing workflows.
- Architected a multi-region model replication service with end-to-end encryption, targeting 99.99% availability across five geographic regions.
- Developed a custom Terraform provider and reusable modules codifying 500+ infrastructure resources, reducing manual configuration by 80%.

- Automated GitHub and SonarQube onboarding using Azure Logic Apps, reducing developer setup from three days to approximately 30 minutes for 200+ developers.
- Contributed \$180K+ in business value through technical consulting, architecture recommendations, and AI/ML solution strategy.

Nagarro, Gurgaon

June 2014 – February 2021

Associate Staff Engineer

- Designed and implemented enterprise backend and search platforms, owning technical design, integration, performance, and delivery across multiple client engagements.
- Led architecture of an enterprise search platform serving 10M+ monthly users, supporting 100K+ concurrent searches with sub-100ms latency and 99.9% availability.
- Designed connectors and synchronization services integrating 15+ heterogeneous data sources into a unified search platform.
- Improved search relevance by 20–25% using ML-based ranking, semantic search, embeddings, and recommendation techniques.
- Designed conversational backend solutions using Microsoft Bot Framework and Google Dialogflow, supporting 50K+ customer interactions monthly with an 85% resolution rate.
- Automated data processing and orchestration workflows, reducing manual effort by 70% and improving data freshness to near real time.
- Mentored engineers on backend engineering, unit testing, cloud-native architecture, ML fundamentals, and engineering best practices.

Amdocs India Limited, Pune

August 2012 – April 2014

Subject Matter Expert

- Implemented enterprise SharePoint Search and backend integration components for portal platforms.
- Developed WCF Windows Services to expose portal data to downstream consumers and integrated enterprise web portals with backend services.

SELECTED ARCHITECTURE & ENGINEERING WORK

Enterprise Event Streaming SDKs – .NET & Python / Confluent Kafka

- Designed reusable SDKs for consuming Confluent Kafka platform APIs with OAuth2, standardizing authentication, integration, error handling, and operational behavior for backend consumers.
- Addressed reliability and disaster-recovery requirements including active-cluster switching and resilient event-stream processing.

Enterprise Search & Distributed Backend Platform

- Architected distributed search services and connectors for 15+ heterogeneous data sources, supporting 10M+ monthly users and 100K+ concurrent searches.
- Used Elasticsearch, caching, semantic retrieval, ML ranking, and asynchronous synchronization pipelines to achieve sub-100ms search latency and 99.9% availability.

Cross-Organization Information Analysis Platform

- Architected an integration platform ingesting data from 20+ APIs and processing 500K+ records daily using Apache Airflow, Azure Data Factory, and Databricks.
- Implemented data-quality, transformation, RBAC, and multilingual processing capabilities for 500+ users across five business units.

Document Intelligence Backend Platform

- Built backend processing pipelines for unstructured documents using AWS Textract, NLP, and SageMaker, processing 10K+ documents in the project and reducing manual reporting effort by 85%.
- Implemented automated training, validation, deployment, monitoring, and data-quality controls, reducing time-to-production from two weeks to one day.

Enterprise Automation & Developer Platform

- Engineered reusable infrastructure and CI/CD patterns with Terraform, GitHub Actions, and Azure DevOps for repeatable cloud provisioning and software delivery.
- Created standardized project templates and engineering documentation adopted across teams, reducing developer onboarding from two weeks to three days.

TECHNICAL SKILLS

Languages: C#, .NET, Python, Go, Java, C++, Node.js, TypeScript, JavaScript, SQL

Backend: ASP.NET Core, Web API, REST, microservices, WCF, FastAPI, Express.js, API design, SDK development, asynchronous processing

Messaging: Apache Kafka, Confluent Cloud, event-driven architecture, streaming, consumer groups, OAuth2 integrations

Cloud: Azure, AWS, GCP; App Service, Functions, Azure ML, Data Factory, SageMaker, EC2, S3, Lambda, RDS, DynamoDB, CloudWatch

Cloud Native: Docker, Kubernetes, AKS, EKS, Helm, Docker Compose, ArgoCD, GitOps, Istio

Infrastructure: Terraform, Bicep, ARM Templates, CloudFormation, Ansible, Azure DevOps, GitHub Actions, GitLab CI

Databases: SQL Server, PostgreSQL, Oracle, MongoDB, Cosmos DB, DynamoDB, Redis, Elasticsearch

Observability: OpenTelemetry, Prometheus, Grafana, Jaeger, New Relic, ELK Stack, distributed tracing, metrics, structured logging

Security: OAuth2, JWT, SAML 2.0, SSO, Azure AD, MFA, RBAC, TLS, identity and access management

AI/ML: AWS SageMaker, Azure ML, Databricks, TensorFlow, PyTorch, Scikit-learn, Hugging Face, LangChain, LangGraph, RAG, embeddings, AI Agents, MCP

Data Engineering: Apache Airflow, Apache Spark, PySpark, Azure Data Factory, Databricks, Pandas, NumPy, Delta Lake

PRINCIPAL-LEVEL LEADERSHIP

- Architecture Ownership:** System design, architecture reviews, technical strategy, API contracts, non-functional requirements, scalability, resilience, security, and production readiness.
- Technical Leadership:** Cross-team technical direction, engineering standards, reusable platform capabilities, mentoring, design reviews, and complex problem solving.
- Platform Thinking:** Build reusable capabilities rather than one-off applications; standardize infrastructure, APIs, integrations, observability, security, and developer workflows.
- Engineering Excellence:** Automation-first delivery, measurable reliability, performance optimization, operational simplicity, disaster recovery, and continuous improvement.

SELECTED IMPACT

- Scale:** Architected backend/search systems serving 10M+ users with sub-100ms latency and 99.9%+ availability.
- Streaming:** Supported 10M+ real-time events daily through Kafka-based enterprise integration.
- Automation:** Reduced operational effort by 97%, from approximately 9 hours to 5 minutes.
- Cloud Platform:** Automated provisioning across 50+ AWS accounts and codified 500+ infrastructure resources.
- Business Value:** Contributed \$180K+ through architecture and technology consulting initiatives.
- AI Enablement:** Applied ML/LLM capabilities to search, document intelligence, MLOps/LLMOps, observability, and autonomous operational workflows.

EDUCATION

MMU, Ambala, India <i>Bachelor of Technology (B.Tech), Computer Science and Engineering</i>	2008 – 2012
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